

课后作业：运动学验证与分析

环境：Ubuntu + ROS + Gazebo

ROS 工作空间：~/ws_gazebo，功能包：humanoid_sim

基础作业

1.1 正运动学验证

- 启动仿真，使用 `arm_kinematics.py fk` 和 `tf_echo /torso /left_hand` 验证正运动学公式。简要给出 3 组关节角，分别发送关节命令，并提交测试结果表格；
- 分析为什么现在输入 `q2` 为正值时，FK 脚本输出与 `tf_echo` 结果不一致？提交思考结果；
- 针对不一致的问题，你会如何修改代码，请给出修改后的代码局部细节和运行结果截图。

1.2 逆运动学闭环验证

- 使用 `arm_kinematics.py ik` 指定末端目标位置，求解 IK，将输出的 `rostopic pub` 命令发送给机器人，用 `tf_echo` 验证手端是否到达目标，提交测试表格；
- 分析 `ik` 的输入参数，即输入的末端位置和 `torso` 中显示的为什么不一致？（提示：`torso` 坐标系和肩部坐标系）给出思考结果，并分析解决思路。

基础作业提交要求

将以上内容整理为一份 PDF，包含：

- 验证测试表格
- 思考题回答
- Gazebo 机器人姿态截图（至少 1 张）
- `tf_echo` 终端输出截图（至少 2 张）

Answers:

1.1 正运动学验证

(1) 测试结果表格

序号	q1	q2	脚本预计算	tf_echo /torso /left_hand 输出
1	0.8	-0.9	x=0.1300, y=0.1255, z=0.0316	0.130, 0.126, 0.032
2	-0.8	-0.9	x=0.1300, y=-0.3220, z=0.2339	0.130, -0.322, 0.233
3	0.0	-0.5	x=0.1300, y=-0.0863, z=-0.0080	0.130, -0.086, -0.008

测试截图：

```
andrew@vml:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/六、机器人运动学基础/作业6.1/workspace$ rosrund humanoid_sim arm_kinematics.py fk 0.0 -0.5
=====
输入: q1 = 0.0000 rad (0.0 deg)
      q2 = -0.5000 rad (-28.6 deg)

手端 (torso 系): x=0.1300, y=-0.0863, z=-0.0080
+ 应与 tf_echo /torso /left_hand 的 Translation 一致

2R 平面坐标: s=-0.0863 (侧向), d=0.3580 (下垂)
=====

>>> 对应的关节命令:
rostopic pub /humanoid/position_controller/command std_msgs/Float64MultiArray "data: [0.0000, 0.0, -0.5000, 0.0, 0.0, 0.0, 0.0, 0.0]" -1
andrew@vml:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/六、机器人运动学基础/作业6.1/workspace$ rostopic pub /humanoid/position_controller/command std_msgs/Float64MultiArray "data: [0.0000, 0.0, -0.5000, 0.0, 0.0, 0.0, 0.0, 0.0]" -1

publishing and latching message for 3.0 seconds
andrew@vml:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/六、机器人运动学基础/作业6.1/workspace$

At time 2152.556
- Translation: [0.130, -0.086, -0.008]
- Rotation: in Quaternion [-0.247, 0.000, 0.000, 0.969]
            in RPY (radian) [-0.499, 0.000, 0.000]
            in RPY (degree) [-28.587, 0.000, 0.000]
At time 2153.576
- Translation: [0.130, -0.087, -0.008]
- Rotation: in Quaternion [-0.248, 0.000, 0.000, 0.969]
            in RPY (radian) [-0.501, 0.000, 0.000]
            in RPY (degree) [-28.706, 0.000, 0.000]
At time 2154.576
- Translation: [0.130, -0.086, -0.008]
- Rotation: in Quaternion [-0.247, 0.000, 0.000, 0.969]
            in RPY (radian) [-0.498, 0.000, 0.000]
            in RPY (degree) [-28.549, 0.000, 0.000]
At time 2155.576
- Translation: [0.130, -0.087, -0.008]
- Rotation: in Quaternion [-0.248, 0.000, 0.000, 0.969]
            in RPY (radian) [-0.502, 0.000, 0.000]
            in RPY (degree) [-28.739, 0.000, 0.000]
At time 2156.576
- Translation: [0.130, -0.086, -0.008]
- Rotation: in Quaternion [-0.247, 0.000, 0.000, 0.969]
            in RPY (radian) [-0.499, 0.000, 0.000]
            in RPY (degree) [-28.592, 0.000, 0.000]
At time 2157.576
- Translation: [0.130, -0.086, -0.008]
- Rotation: in Quaternion [-0.247, 0.000, 0.000, 0.969]
            in RPY (radian) [-0.499, 0.000, 0.000]
            in RPY (degree) [-28.569, 0.000, 0.000]
At time 2158.576
- Translation: [0.130, -0.087, -0.008]
- Rotation: in Quaternion [-0.248, 0.000, 0.000, 0.969]
            in RPY (radian) [-0.502, 0.000, 0.000]
            in RPY (degree) [-28.736, 0.000, 0.000]
At time 2159.576
- Translation: [0.130, -0.086, -0.008]
```

(2) 思考题

q2 代表左肘关节转动角，在 urdf 文件中限制范围[-2.0, 0]。当 q2 输入正值后，会被修正为 0，也就是相对于左上臂无转动，导致 FK 脚本输出与 tf_echo 结果不一致。

```
<joint name="left_elbow" type="revolute">
  <parent link="left_upper_arm"/>
  <child link="left_forearm"/>
  <origin xyz="0 0 -0.2" rpy="0 0 0"/>
  <axis xyz="1 0 0"/>
  <limit lower="-2.0" upper="0" effort="10" velocity="1.0"/>
</joint>
```

可以修改 urdf 中的 limit 属性或者完善 arm_kinematics.py。

1. 修改 urdf 中的 limit 属性

```
<limit lower="-2.0" upper="2.0" effort="10" velocity="1.0"/>
```

运行结果：

```
andrew@vml:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/六、机器人运动学基础/作业6.1/workspace$ rosrund humanoid_sim arm_kinematics.py fk 0.5 0.5
=====
输入: q1 = 0.5000 rad (28.6 deg)
      q2 = 0.5000 rad (28.6 deg)

手端 (torso 系): x=0.1300, y=0.2473, z=0.0772
+ 应与 tf_echo /torso /left_hand 的 Translation 一致

2R 平面坐标: s=0.2473 (侧向), d=0.2728 (下垂)
=====

>>> 对应的关节命令:
rostopic pub /humanoid/position_controller/command std_msgs/Float64MultiArray "data: [0.5000, 0.0, 0.5000, 0.0, 0.0, 0.0, 0.0, 0.0]" -1
andrew@vml:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/六、机器人运动学基础/作业6.1/workspace$ rostopic pub /humanoid/position_controller/command std_msgs/Float64MultiArray "data: [0.5000, 0.0, 0.5000, 0.0, 0.0, 0.0, 0.0, 0.0]" -1
publishing and latching message for 3.0 seconds
```

```
At time 115.714
- Translation: [0.130, 0.247, 0.077]
- Rotation: in Quaternion [0.478, 0.000, 0.000, 0.878]
           in RPY (radian) [0.998, -0.000, 0.000]
           in RPY (degree) [57.156, -0.000, 0.000]
At time 116.714
- Translation: [0.130, 0.247, 0.077]
- Rotation: in Quaternion [0.479, 0.000, 0.000, 0.878]
           in RPY (radian) [0.998, -0.000, 0.000]
           in RPY (degree) [57.188, -0.000, 0.000]
At time 117.714
- Translation: [0.130, 0.248, 0.078]
- Rotation: in Quaternion [0.480, 0.000, 0.000, 0.877]
           in RPY (radian) [1.002, -0.000, 0.000]
           in RPY (degree) [57.424, -0.000, 0.000]
At time 118.714
- Translation: [0.130, 0.248, 0.077]
- Rotation: in Quaternion [0.480, 0.000, 0.000, 0.877]
           in RPY (radian) [1.001, -0.000, 0.000]
           in RPY (degree) [57.343, -0.000, 0.000]
At time 119.714
- Translation: [0.130, 0.247, 0.077]
- Rotation: in Quaternion [0.479, 0.000, 0.000, 0.878]
           in RPY (radian) [1.000, -0.000, 0.000]
           in RPY (degree) [57.287, -0.000, 0.000]
At time 120.714
- Translation: [0.130, 0.247, 0.077]
- Rotation: in Quaternion [0.479, 0.000, 0.000, 0.878]
           in RPY (radian) [0.999, -0.000, 0.000]
           in RPY (degree) [57.226, -0.000, 0.000]
```

2. 完善 arm_kinematics.py

```
# ===== fk 子命令 =====
```

```

def cmd_fk(args):
    q1 = float(args[0]) if len(args) > 0 else 0.8
    q2 = float(args[1]) if len(args) > 1 else -0.9
    q1_limited, q2_limited = False, False
    # 限位判断
    if q1 < Q1_LIMITS[0]:
        q1 = Q1_LIMITS[0]
        q1_limited = True
    elif q1 > Q1_LIMITS[1]:
        q1 = Q1_LIMITS[1]
        q1_limited = True

    if q2 < Q2_LIMITS[0]:
        q2 = Q2_LIMITS[0]
        q2_limited = True
    elif q2 > Q2_LIMITS[1]:
        q2 = Q2_LIMITS[1]
        q2_limited = True

    s, d = fk_2r(q1, q2)
    x, y, z = to_torso(s, d)

    print("=" * 55)
    print(f"输入: q1 = {q1:.4f} rad ({math.degrees(q1):.1f} deg) {'限位'
if q1_limited else ''}")
    print(f"      q2 = {q2:.4f} rad ({math.degrees(q2):.1f} deg) {'限位'
if q2_limited else ''}")

```

运行结果:

```

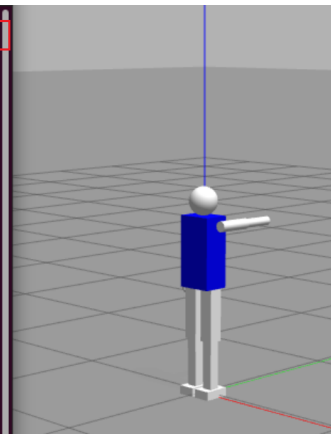
andrew@vml:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/六、机器人运动学基础/作业6.1/workspace$ rosrn humanoid_sim arm_kinematics.py fk 2.5 0.5
=====
输入: q1 = 1.5700 rad (90.0 deg) 限位
      q2 = 0.0000 rad (0.0 deg) 限位

手端 (torso 系): x=0.1300, y=0.3800, z=0.3497
               : 应与 tf_echo /torso /left_hand 的 translation 一致

2R 平面坐标: s=0.3800 (侧向), d=0.0003 (下垂)
=====

>>> 对应的关节命令:
rostopic pub /humanoid/position_controller/command std_msgs/Float64MultiArray "data: [1.5700, 0.0, 0.0000, 0.0, 0.0, 0.0, 0.0, 0.0]" -1
andrew@vml:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/六、机器人运动学基础/作业6.1/workspace$ rostopic pub /humanoid/position_controller/command std_msgs/Float64MultiArray "data: [1.5700, 0.0, 0.0000, 0.0, 0.0, 0.0, 0.0, 0.0]" -1
publishing and latching message for 3.0 seconds
andrew@vml:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/六、机器人运动学基础/作业6.1/workspace$

```



```

At time 375.979
- Translation: [0.130, 0.380, 0.349]
- Rotation: in Quaternion [0.706, 0.000, 0.000, 0.708]
            in RPY (radian) [1.567, -0.000, 0.000]
            in RPY (degree) [89.793, -0.000, 0.000]
At time 376.979
- Translation: [0.130, 0.380, 0.348]
- Rotation: in Quaternion [0.705, 0.000, 0.000, 0.709]
            in RPY (radian) [1.566, -0.000, 0.000]
            in RPY (degree) [89.714, -0.000, 0.000]
At time 377.979
- Translation: [0.130, 0.380, 0.349]
- Rotation: in Quaternion [0.706, 0.000, 0.000, 0.708]
            in RPY (radian) [1.567, -0.000, 0.000]
            in RPY (degree) [89.809, -0.000, 0.000]
At time 378.979
- Translation: [0.130, 0.380, 0.349]
- Rotation: in Quaternion [0.706, 0.000, 0.000, 0.709]
            in RPY (radian) [1.567, -0.000, 0.000]
            in RPY (degree) [89.765, -0.000, 0.000]
At time 379.979
- Translation: [0.130, 0.380, 0.349]
- Rotation: in Quaternion [0.706, 0.000, 0.000, 0.708]
            in RPY (radian) [1.568, -0.000, 0.000]
            in RPY (degree) [89.822, -0.000, 0.000]

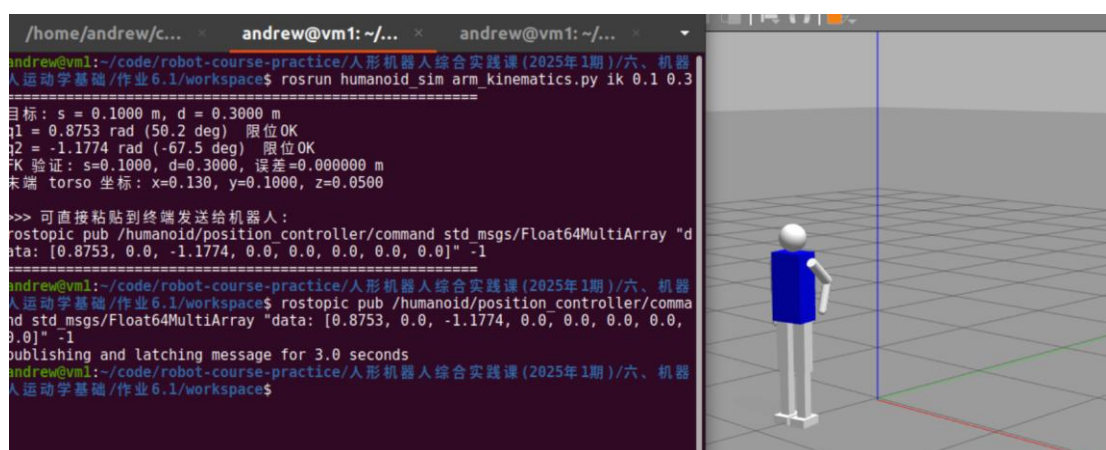
```

1.2 逆运动学闭环验证

(1) 测试结果表格

序号	s	d	脚本预计算	tf_echo /torso /left_hand 输出
1	0.1	0.3	x=0.130, y=0.1000, z=0.0500	0.130, 0.100, 0.050
2	0.2	0.2	x=0.130, y=0.2000, z=0.1500	0.130, 0.200, 0.150
3	0.2	0.3	x=0.130, y=0.2000, z=0.0500	0.130, 0.199, 0.050

测试截图：




```

At time 768.779
- Translation: [0.130, 0.000, -0.030]
- Rotation: in Quaternion [0.000, 0.000, 0.000, 1.000]
            in RPY (radian) [0.000, -0.000, 0.000]
            in RPY (degree) [0.024, -0.000, 0.000]
At time 768.779
- Translation: [0.130, 0.000, -0.030]
- Rotation: in Quaternion [0.000, 0.000, 0.000, 1.000]
            in RPY (radian) [0.000, -0.000, 0.000]
            in RPY (degree) [0.024, -0.000, 0.000]
At time 768.779
- Translation: [0.130, 0.000, -0.030]
- Rotation: in Quaternion [0.000, 0.000, 0.000, 1.000]
            in RPY (radian) [0.000, -0.000, 0.000]
            in RPY (degree) [0.024, -0.000, 0.000]
At time 768.779
- Translation: [0.130, 0.000, -0.030]
- Rotation: in Quaternion [0.000, 0.000, 0.000, 1.000]
            in RPY (radian) [0.000, -0.000, 0.000]
            in RPY (degree) [0.024, -0.000, 0.000]
At time 768.779
- Translation: [0.130, 0.000, -0.030]
- Rotation: in Quaternion [0.000, 0.000, 0.000, 1.000]
            in RPY (radian) [0.000, -0.000, 0.000]
            in RPY (degree) [0.024, -0.000, 0.000]

```



(2) 思考题

ik 的输入参数为 s 、 d ，位于肩部坐标系的 yz 二维平面内。肩部坐标系同 torso 坐标系需要通过 x 、 z 轴平移达到，也就导致输入的末端位置和 torso 中显示的并不一致。

```

andrew@vm1:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/六、机器人运动学基础/作业6.1/workspace$ rosrn humanoid_sim arm_kinematics.py ik 0.2 0.3
=====
目标: s = 0.2000 m, d = 0.3000 m
q1 = 0.8922 rad (51.1 deg) 限位OK
q2 = -0.6435 rad (-36.9 deg) 限位OK
FK 验证: s=0.2000, d=0.3000, 误差=0.000000 m
末端 torso 坐标: x=0.130, y=0.2000, z=0.0500

>>> 可直接粘贴到终端发送给机器人:
rostopic pub /humanoid/position_controller/command std_msgs/Float64MultiArray "d
ata: [0.8922, 0.0, -0.6435, 0.0, 0.0, 0.0, 0.0, 0.0]" -1
=====

```

解决思路：根据 urdf 文件找出肩部坐标系同 torso 坐标系的差异、变换矩阵，将 (s, d) 转换成 torso 坐标系下的值。

```
SHOULDER_IN_TORSO = (0.13, 0.0, 0.35)
Q1_LIMITS = (-1.57, 1.57)
Q2_LIMITS = (-2.0, 0.0)
RMAX = L1 + L2
RMIN = abs(L1 - L2)

def fk_2r(q1, q2):
    """2R 平面 FK → (s, d)"""
    s = L1 * math.sin(q1) + L2 * math.sin(q1 + q2)
    d = L1 * math.cos(q1) + L2 * math.cos(q1 + q2)
    return s, d

def to_torso(s, d):
    """(s, d) → torso 坐标系 (x, y, z)"""
    return SHOULDER_IN_TORSO[0], s, SHOULDER_IN_TORSO[2] - d
```